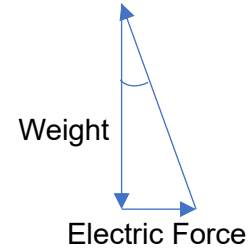


4	(a)		Coulomb's law states that the magnitude of <u>the electric force between two point charges is directly proportional to the product of the magnitude of the charges and inversely proportional to the square of their separation.</u>	B1
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	(b)	(i)	Electric force, $F_E = \frac{96 \times 10^{-9} \times Q_Y}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.080^2}$ angle to the vertical, $\theta = \sin^{-1}\left(\frac{0.080}{1.2}\right) (= 3.823^\circ)$ $F_E = mg \tan \theta$ $= (0.013)(9.81) \tan(3.823^\circ)$ OR $= (0.013)(9.81) \left[\frac{0.080}{\sqrt{1.2^2 - 0.080^2}} \right]$ $\frac{96 \times 10^{-9} \times Q_Y}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.080^2} = (0.013)(9.81) \tan(3.823^\circ)$ $Q_Y = 63 \text{ nC (shown)}$	M1 M1 A0
		(ii)	$\phi_{\text{net}} = \frac{96 \times 10^{-9}}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.040} + \frac{63 \times 10^{-9}}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.040}$ $= 21580 + 14162$ $= 3.57 \times 10^4 \text{ V}$	C1 A1
	(c)	(i)	$E = \frac{V}{d} = \frac{250}{0.018} (= 1.389 \times 10^4 \text{ V m}^{-1})$ $a = \frac{F}{m} = \frac{qE}{m} = \frac{(1.6 \times 10^{-19})(1.389 \times 10^4)}{(9.11 \times 10^{-31})}$ $= 2.4 \times 10^{15} \text{ m s}^{-2} \text{ (2sf)}$	M1 M1 A0
		(ii)	time taken to travel through the parallel plates = length/ speed $= \frac{20 \text{ cm}}{2.0 \times 10^7} (= 1.0 \times 10^{-8} \text{ s})$ final vertical speed $= u_y + a_y t = 0 + \frac{F}{m} t$ $= (2.4 \times 10^{15})(1.0 \times 10^{-8})$ $(= 2.4 \times 10^7 \text{ m s}^{-1})$ speed $= \sqrt{(2.0 \times 10^7)^2 + (2.4 \times 10^7)^2}$ $= 3.1 \times 10^7 \text{ m s}^{-1}$	C1 C1 A1
		(iii)	The <u>polarity of the charge is different</u>, so the force will be in the opposite direction to that of the electron. Hence, the proton will be <u>deflected down</u> instead of up. The <u>mass of the proton is heavier</u> than that of the electron, so the acceleration is lower. Hence, the <u>deflection will be less</u> than that of the electron.	B1 B1



5	(a)	(i)	Resistance is infinite / very high	B1
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